

Aadhaar Migration Analytics – UIDAI Data Hackathon 2026

Prepared by: Aryan Singh

1. Problem Statement

Internal migration in India is challenging to measure in real time. Aadhaar update patterns offer a live signal of population mobility. This analysis extracts migration flows using UIDAI enrolment and update data.

2. Objectives

- Identify migration-receiving states and districts
- Identify migration-sending regions
- Separate job-driven vs family migration
- Detect seasonal migration peaks
- Forecast Aadhaar update demand
- Recommend UIDAI resource planning actions

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
from prophet import Prophet
```

```
In [3]: df_enrol = pd.read_csv(r"D:\UIDAI COMPETITION\clean_data\enrolment_merged.csv")
df_demo = pd.read_csv(r"D:\UIDAI COMPETITION\clean_data\demographic_merged.csv")
df_bio = pd.read_csv(r"D:\UIDAI COMPETITION\clean_data\biometric_merged.csv")

print("Data Loaded")
```

Data Loaded

```
In [4]: for df in [df_enrol, df_demo, df_bio]:
df['date'] = pd.to_datetime(df['date'], dayfirst=True)
df['month'] = df['date'].dt.to_period('M')
```

3. Methodology

- Load and merge files
- Convert dates to monthly periods
- Aggregate demographic and enrolment values
- Create migration index
- Compare state and district levels

- Segment by age
- Analyse season trends
- Forecast demand

```
In [5]: enrol_month = df_enrol.groupby(['state', 'district', 'month'])[['age_0_5', 'age_5_17',
enrol_month['total_enrol'] = enrol_month[['age_0_5', 'age_5_17', 'age_18_greater']].sum(axis=1)

demo_month = df_demo.groupby(['state', 'district', 'month'])[['demo_age_5_17', 'demo_age_17_18',
demo_month['total_demo'] = demo_month[['demo_age_5_17', 'demo_age_17_18']].sum(axis=1)
```

```
In [6]: migration_df = demo_month.merge(
    enrol_month[['state', 'district', 'month', 'total_enrol']],
    on=['state', 'district', 'month'],
    how='left'
)

migration_df['migration_index'] = migration_df['total_demo'] / migration_df['total_enrol']
```

4. State-Level Migration Analysis

```
In [7]: state_summary = migration_df.groupby('state')['migration_index'].mean().reset_index()

top_receivers = state_summary.sort_values('migration_index', ascending=False).head(10)
top_senders = state_summary.sort_values('migration_index').head(10)

top_receivers
```

```
Out[7]:
```

	state	migration_index
14	Daman & Diu	40.708333
17	Delhi	31.734279
33	Manipur	27.046568
32	Maharashtra	25.814889
8	Chandigarh	25.591156
55	WESTBENGAL	21.000000
61	Westbengal	19.500000
7	Bihar	18.389381
46	Rajasthan	18.097513
13	Dadra and Nagar Haveli and Daman and Diu	17.592596

```
In [9]: import matplotlib.pyplot as plt

plt.figure(figsize=(12,6))
plt.style.use("seaborn-v0_8-darkgrid")
```

```

bars = plt.bar(
    top_receivers['state'],
    top_receivers['migration_index'],
    color='#2E8B57',
    edgecolor='black'
)

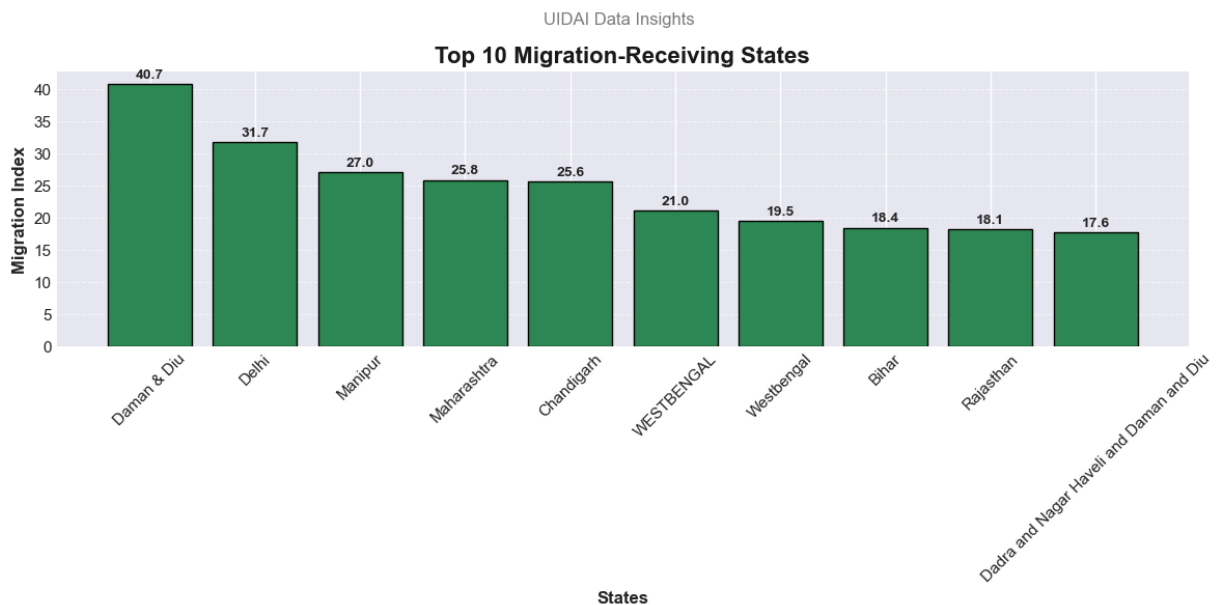
for bar in bars:
    yval = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        yval + 0.5,
        f"{yval:.1f}",
        ha='center', va='bottom', fontsize=10, fontweight='bold'
    )

plt.xticks(rotation=45, fontsize=11, fontweight='medium')
plt.yticks(fontsize=11)
plt.ylabel("Migration Index", fontsize=12, fontweight='bold')
plt.xlabel("States", fontsize=12, fontweight='bold')

plt.title("Top 10 Migration-Receiving States", fontsize=16, fontweight='bold', color='green')
plt.suptitle("UIDAI Data Insights", fontsize=12, color='gray')

plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()

```



```

In [10]: import matplotlib.pyplot as plt

plt.figure(figsize=(12,6))
plt.style.use("seaborn-v0_8-darkgrid")

bars = plt.bar(
    top_senders['state'],
    top_senders['migration_index'],
    color='#B22222',

```

```

    edgecolor='black'
)

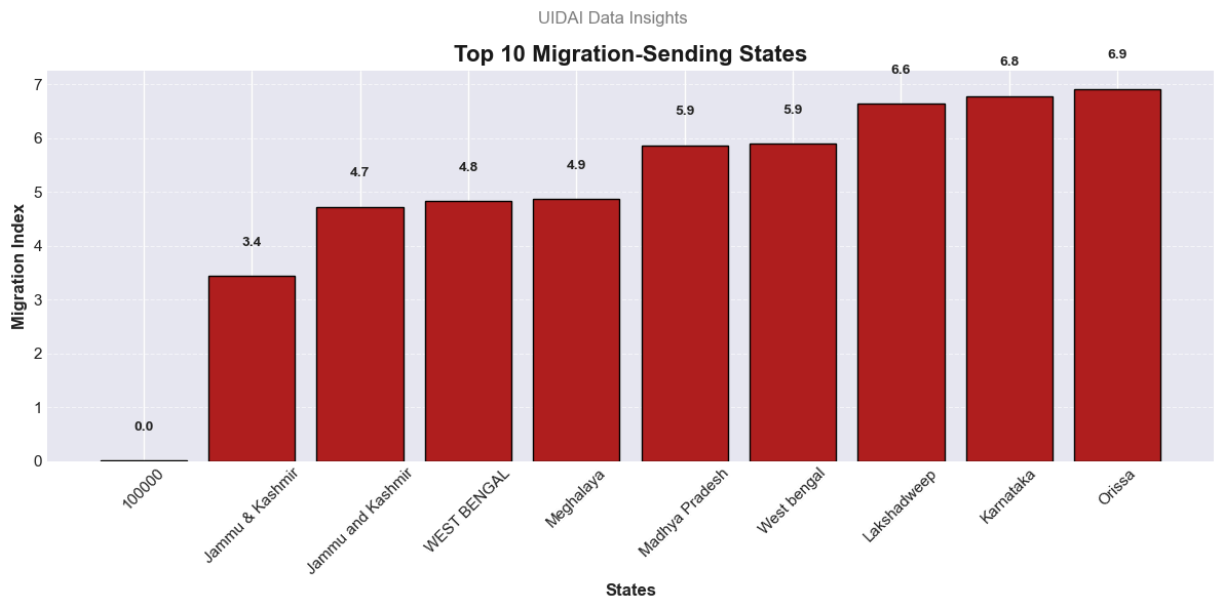
for bar in bars:
    yval = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        yval + 0.5,
        f"{yval:.1f}",
        ha='center', va='bottom', fontsize=10, fontweight='bold'
    )

plt.xticks(rotation=45, fontsize=11, fontweight='medium')
plt.yticks(fontsize=11)
plt.ylabel("Migration Index", fontsize=12, fontweight='bold')
plt.xlabel("States", fontsize=12, fontweight='bold')

plt.title("Top 10 Migration-Sending States", fontsize=16, fontweight='bold', color=
plt.suptitle("UIDAI Data Insights", fontsize=12, color='gray')

plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()

```



5. District-Level Hotspots

```

In [11]: district_summary = migration_df.groupby(['state', 'district'])['migration_index'].me

top_districts = district_summary.sort_values('migration_index', ascending=False).he
bottom_districts = district_summary.sort_values('migration_index').head(20)

top_districts

```

Out[11]:

	state	district	migration_index
749	Rajasthan	Beawar	449.000000
769	Rajasthan	Jalore	384.000000
762	Rajasthan	Didwana-Kuchaman	308.000000
515	Maharashtra	Ahilyanagar	186.000000
563	Maharashtra	Thane	185.716841
934	Uttar Pradesh	Gorakhpur	169.909337
222	Delhi	North West Delhi	149.666466
141	Bihar	Muzaffarpur	135.716791
783	Rajasthan	Salumbar	127.000000
145	Bihar	Patna	126.208701
759	Rajasthan	Deeg	124.916667
123	Bihar	Bhagalpur	105.936942
351	Jharkhand	Deoghar	100.345644
918	Uttar Pradesh	Bulandshahr	85.804291
745	Rajasthan	Balotra	79.000000
991	Uttarakhand	Haridwar	65.680776
392	Karnataka	Bengaluru South	63.331593
270	Gujarat	Vadodara	62.906538
193	Chhattisgarh	Mohla-Manpur-Ambagarh Chouki	61.438366
909	Uttar Pradesh	Banda	60.457465

```
In [12]: import matplotlib.pyplot as plt

plt.figure(figsize=(14,6))
plt.style.use("seaborn-v0_8-darkgrid")

bars = plt.bar(
    top_districts['district'],
    top_districts['migration_index'],
    color='#4682B4',
    edgecolor='black'
)

for bar in bars:
    yval = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        yval + 0.5,
```

```

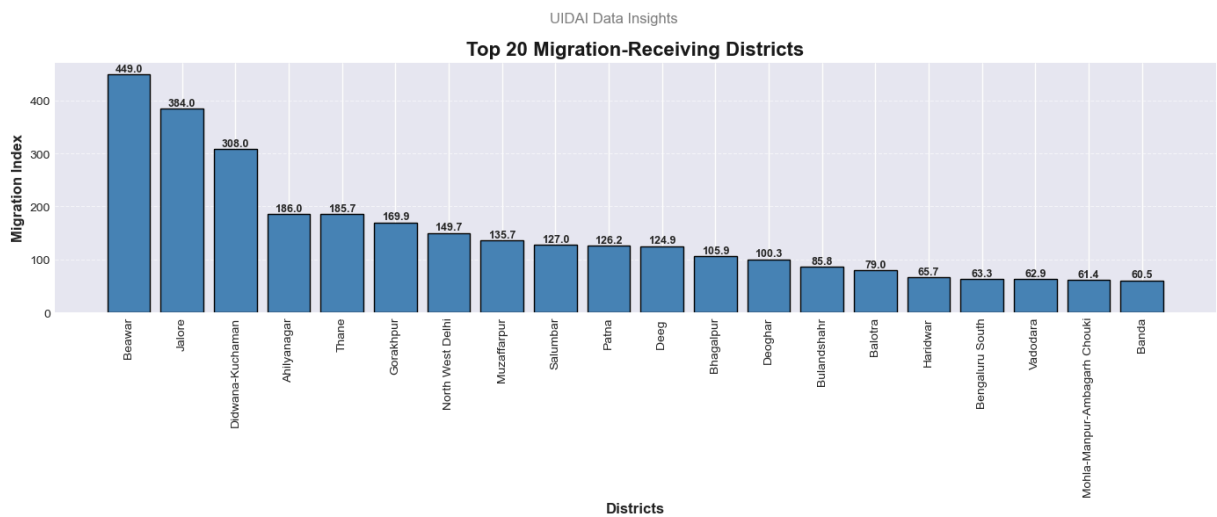
        f"{yval:.1f}",
        ha='center', va='bottom', fontsize=9, fontweight='bold'
    )

plt.xticks(rotation=90, fontsize=10, fontweight='medium')
plt.yticks(fontsize=10)
plt.ylabel("Migration Index", fontsize=12, fontweight='bold')
plt.xlabel("Districts", fontsize=12, fontweight='bold')

plt.title("Top 20 Migration-Receiving Districts", fontsize=16, fontweight='bold', c
plt.suptitle("UIDAI Data Insights", fontsize=12, color='gray')

plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()

```



6. Family vs Job Migration

```

In [13]: migration_df['child_ratio'] = migration_df['demo_age_5_17'] / migration_df['total_d
migration_df['adult_ratio'] = migration_df['demo_age_17_'] / migration_df['total_d

child_states = migration_df.groupby('state')['child_ratio'].mean().reset_index().so
adult_states = migration_df.groupby('state')['adult_ratio'].mean().reset_index().so

child_states

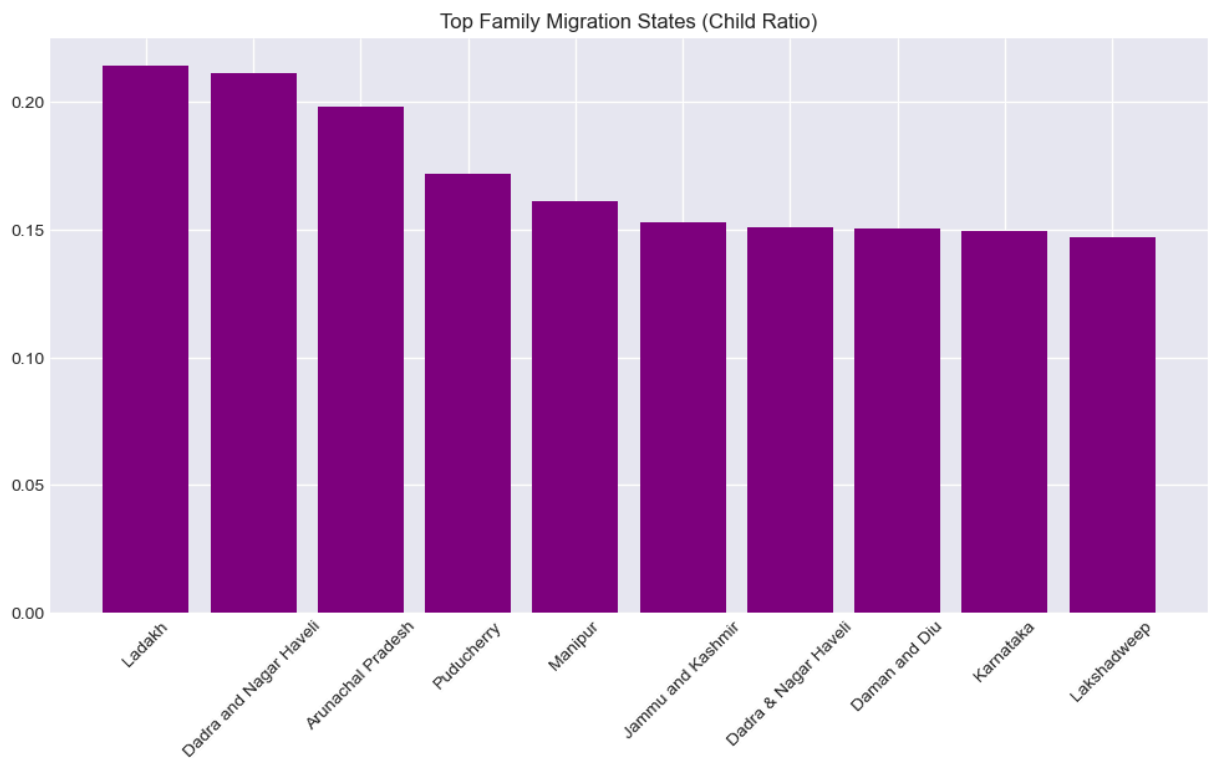
```

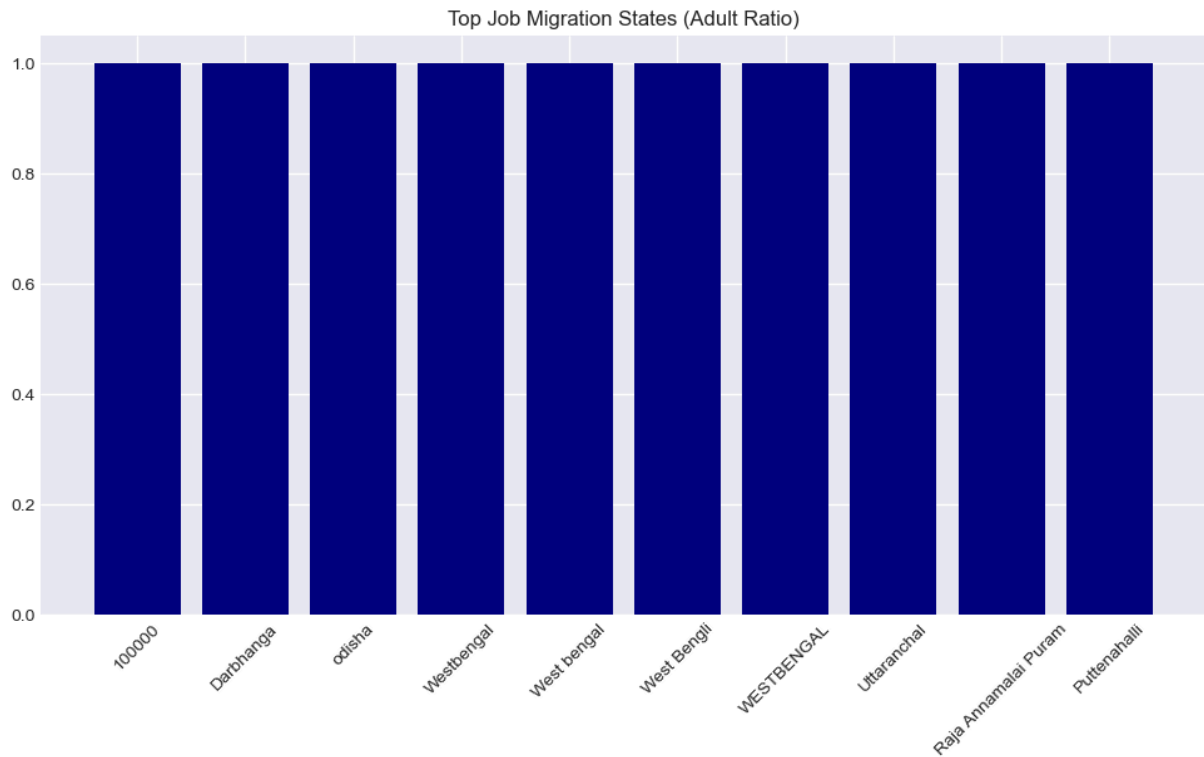
Out[13]:

	state	child_ratio
28	Ladakh	0.214332
12	Dadra and Nagar Haveli	0.211156
4	Arunachal Pradesh	0.198171
42	Puducherry	0.171872
33	Manipur	0.161111
24	Jammu and Kashmir	0.152776
11	Dadra & Nagar Haveli	0.151094
15	Daman and Diu	0.150347
26	Karnataka	0.149538
29	Lakshadweep	0.146983

```
In [15]: plt.figure(figsize=(12,6))
plt.bar(child_states['state'], child_states['child_ratio'], color='purple')
plt.xticks(rotation=45)
plt.title("Top Family Migration States (Child Ratio)")
plt.show()

plt.figure(figsize=(12,6))
plt.bar(adult_states['state'], adult_states['adult_ratio'], color='navy')
plt.xticks(rotation=45)
plt.title("Top Job Migration States (Adult Ratio)")
plt.show()
```

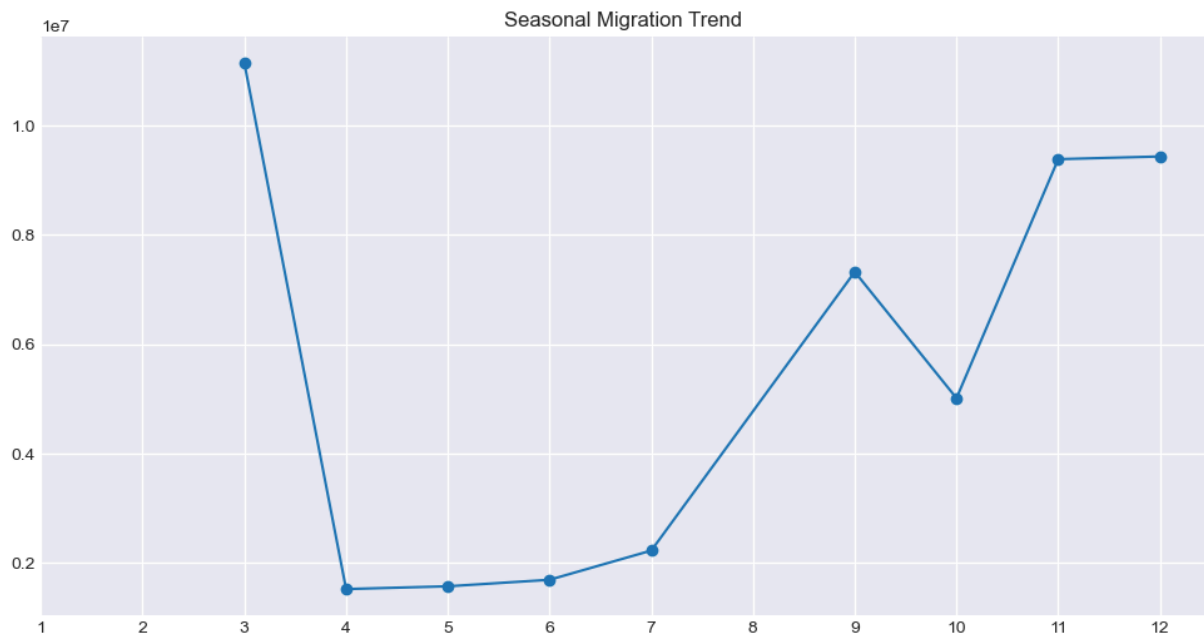




7. Seasonal Migration Pattern

```
In [16]: migration_df['month_num'] = migration_df['month'].dt.month
seasonality = migration_df.groupby('month_num')['total_demo'].sum().reset_index()

plt.figure(figsize=(12,6))
plt.plot(seasonality['month_num'], seasonality['total_demo'], marker='o')
plt.xticks(range(1,13))
plt.title("Seasonal Migration Trend")
plt.show()
```



8. Forecasting Aadhaar Update Demand

```
In [18]: forecast_df = migration_df.groupby('month')['total_demo'].sum().reset_index()
forecast_df['month'] = forecast_df['month'].dt.to_timestamp()
forecast_df.columns = ['ds', 'y']

model = Prophet()
model.fit(forecast_df)
future = model.make_future_dataframe(periods=12, freq='M')
forecast = model.predict(future)

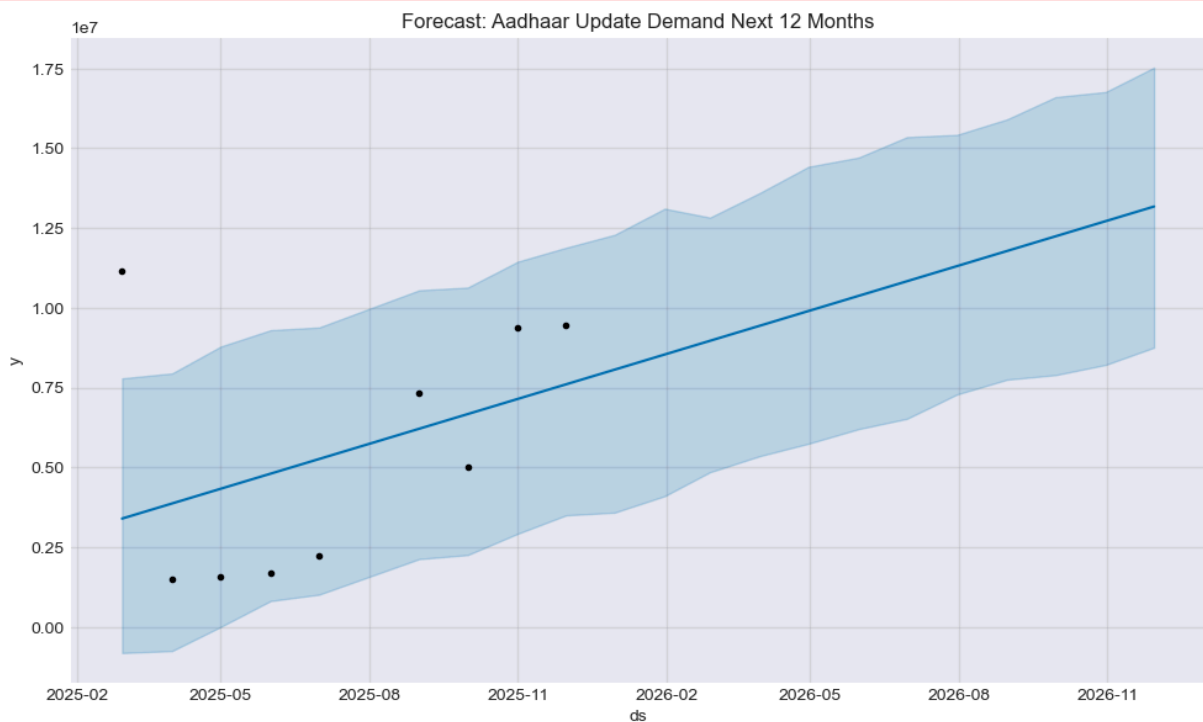
model.plot(forecast)
plt.title("Forecast: Aadhaar Update Demand Next 12 Months")
plt.show()
```

14:17:01 - cmdstanpy - INFO - Chain [1] start processing

14:17:01 - cmdstanpy - INFO - Chain [1] done processing

C:\Users\Aryan\anaconda3\Lib\site-packages\prophet\forecaster.py:1872: FutureWarning: 'M' is deprecated and will be removed in a future version, please use 'ME' instead.

```
dates = pd.date_range(
```



9. Recommendations

- Expand UIDAI update counters in Tamil Nadu, Jharkhand & top districts
- Deploy mobile Aadhaar vans in NE sending states
- Seasonal staffing during April–July migration peaks
- Promote digital self-service update channels

10. Limitations

- Aadhaar updates approximate migration, not exact
- Local mobility within districts cannot be separated
- Missing demographic attributes limits segmentation

11. Future Work

- Add gender-based insights if data becomes available
- Build district-to-district migration corridors
- Compare with railway/bus ticket datasets

12. Conclusion

Aadhaar demographic update data is a powerful proxy for internal migration. This study shows receiving and sending states, hotspot districts, family vs job-driven migration, seasonal peaks and future demand projection. UIDAI can use this as a scalable planning tool for citizen service delivery.

```
In [19]: pip install nbconvert pyppeteer
```

Requirement already satisfied: nbconvert in c:\users\aryan\anaconda3\lib\site-packages (7.16.4)

Collecting pyppeteer

Downloading pyppeteer-2.0.0-py3-none-any.whl.metadata (7.1 kB)

Requirement already satisfied: beautifulsoup4 in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (4.12.3)

Requirement already satisfied: bleach!=5.0.0 in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (4.1.0)

Requirement already satisfied: defusedxml in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (0.7.1)

Requirement already satisfied: Jinja2>=3.0 in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (3.1.4)

Requirement already satisfied: jupyter-core>=4.7 in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (5.7.2)

Requirement already satisfied: jupyterlab-pygments in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (0.1.2)

Requirement already satisfied: MarkupSafe>=2.0 in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (2.1.3)

Requirement already satisfied: mistune<4,>=2.0.3 in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (2.0.4)

Requirement already satisfied: nbclient>=0.5.0 in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (0.8.0)

Requirement already satisfied: nbformat>=5.7 in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (5.10.4)

Requirement already satisfied: packaging in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (24.1)

Requirement already satisfied: pandocfilters>=1.4.1 in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (1.5.0)

Requirement already satisfied: pygments>=2.4.1 in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (2.15.1)

Requirement already satisfied: tinycss2 in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (1.2.1)

Requirement already satisfied: traitlets>=5.1 in c:\users\aryan\anaconda3\lib\site-packages (from nbconvert) (5.14.3)

Requirement already satisfied: appdirs<2.0.0,>=1.4.3 in c:\users\aryan\anaconda3\lib\site-packages (from pyppeteer) (1.4.4)

Requirement already satisfied: certifi>=2023 in c:\users\aryan\anaconda3\lib\site-packages (from pyppeteer) (2024.8.30)

Requirement already satisfied: importlib-metadata>=1.4 in c:\users\aryan\anaconda3\lib\site-packages (from pyppeteer) (7.0.1)

Collecting pyee<12.0.0,>=11.0.0 (from pyppeteer)

Downloading pyee-11.1.1-py3-none-any.whl.metadata (2.8 kB)

Requirement already satisfied: tqdm<5.0.0,>=4.42.1 in c:\users\aryan\anaconda3\lib\site-packages (from pyppeteer) (4.66.5)

Collecting urllib3<2.0.0,>=1.25.8 (from pyppeteer)

Downloading urllib3-1.26.20-py2.py3-none-any.whl.metadata (50 kB)

Collecting websockets<11.0,>=10.0 (from pyppeteer)

Downloading websockets-10.4.tar.gz (84 kB)

Preparing metadata (setup.py): started

Preparing metadata (setup.py): finished with status 'done'

Requirement already satisfied: six>=1.9.0 in c:\users\aryan\anaconda3\lib\site-packages (from bleach!=5.0.0->nbconvert) (1.16.0)

Requirement already satisfied: webencodings in c:\users\aryan\anaconda3\lib\site-packages (from bleach!=5.0.0->nbconvert) (0.5.1)

Requirement already satisfied: zipp>=0.5 in c:\users\aryan\anaconda3\lib\site-packages (from importlib-metadata>=1.4->pyppeteer) (3.17.0)

Requirement already satisfied: platformdirs>=2.5 in c:\users\aryan\anaconda3\lib\site-packages (from jupyter-core>=4.7->nbconvert) (3.10.0)

Requirement already satisfied: pywin32>=300 in c:\users\aryan\anaconda3\lib\site-packages (from jupyter-core>=4.7->nbconvert) (305.1)

Requirement already satisfied: jupyter-client>=6.1.12 in c:\users\aryan\anaconda3\lib\site-packages (from nbclient>=0.5.0->nbconvert) (8.6.0)

Requirement already satisfied: fastjsonschema>=2.15 in c:\users\aryan\anaconda3\lib\site-packages (from nbformat>=5.7->nbconvert) (2.16.2)

Requirement already satisfied: jsonschema>=2.6 in c:\users\aryan\anaconda3\lib\site-packages (from nbformat>=5.7->nbconvert) (4.23.0)

Requirement already satisfied: typing-extensions in c:\users\aryan\anaconda3\lib\site-packages (from pyee<12.0.0,>=11.0.0->pypeteer) (4.11.0)

Requirement already satisfied: colorama in c:\users\aryan\anaconda3\lib\site-packages (from tqdm<5.0.0,>=4.42.1->pypeteer) (0.4.6)

Requirement already satisfied: soupsieve>1.2 in c:\users\aryan\anaconda3\lib\site-packages (from beautifulsoup4->nbconvert) (2.5)

Requirement already satisfied: attrs>=22.2.0 in c:\users\aryan\anaconda3\lib\site-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (23.1.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in c:\users\aryan\anaconda3\lib\site-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (2023.7.1)

Requirement already satisfied: referencing>=0.28.4 in c:\users\aryan\anaconda3\lib\site-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.30.2)

Requirement already satisfied: rpds-py>=0.7.1 in c:\users\aryan\anaconda3\lib\site-packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.10.6)

Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\aryan\anaconda3\lib\site-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (2.9.0.post0)

Requirement already satisfied: pyzmq>=23.0 in c:\users\aryan\anaconda3\lib\site-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (25.1.2)

Requirement already satisfied: tornado>=6.2 in c:\users\aryan\anaconda3\lib\site-packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (6.4.1)

Downloading pypeteer-2.0.0-py3-none-any.whl (82 kB)

Downloading pyee-11.1.1-py3-none-any.whl (15 kB)

Downloading urllib3-1.26.20-py2.py3-none-any.whl (144 kB)

Building wheels for collected packages: websockets

Building wheel for websockets (setup.py): started

Building wheel for websockets (setup.py): finished with status 'done'

Created wheel for websockets: filename=websockets-10.4-cp312-cp312-win_amd64.whl size=95034 sha256=3d1f3b75fab51ac67af6e9fb77646a6e12c1a87a2656bf4c1c6224cfc2154471

Stored in directory: c:\users\aryan\appdata\local\pip\cache\wheels\80\cf\6d\5d7e4c920cb41925a178b2d2621889c520d648bab487b1d7fd

Successfully built websockets

Installing collected packages: websockets, urllib3, pyee, pypeteer

Attempting uninstall: urllib3

Found existing installation: urllib3 2.2.3

Uninstalling urllib3-2.2.3:

Successfully uninstalled urllib3-2.2.3

Successfully installed pyee-11.1.1 pypeteer-2.0.0 urllib3-1.26.20 websockets-10.4

Note: you may need to restart the kernel to use updated packages.

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

conda-repo-cli 1.0.114 requires urllib3>=2.2.2, but you have urllib3 1.26.20 which is incompatible.

